

HARI SARAVANAN D

Embedded Software Engineer

CONTACT DETAILS

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EDUCATION

B.E. IN MECHATRONICS ENGINEERING

06/2016 - 04/2020

KUMARAGURU COLLEGE OF TECHNOLOGY,
COIMBATORE

SKILLS

Programming Languages:

- Embedded C, C++, Python, Shell Scripting, Object-Oriented Design (OOAD)

Embedded Systems & Microcontrollers:

- ARM Cortex-M, STM32, ESP32
- RTOS, Bare-Metal Programming, Bootloader, Embedded Linux, xilinx zynq ultrascale+ mpsoc, Yocto, Buildroot System, Secure Boot

Hardware Interaction:

- GPIOs, Timers, ADC, UART, SPI, I2C, Watch Dog, PWM, Timers

Development Tools:

- Keil, STM32Cube, Git version control, CMake

Debugging Tools:

- JTAG, ST-Link, Oscilloscope, Logic Analyzer, Multimeter, Serial log

Communication Protocols:

- Modbus-RTU, MQTT, LWIP Stack

Other Skills:

- Reading datasheets and schematics
- Firmware Architecture Design
- Firmware development life cycle
- Agile/Scrum Methodologies
- Structured Problem Solving
- Board Bring-up
- Cross Functional Team Collaboration

LANGUAGES

- English : Fluent
- Tamil : Native

PROFILE SUMMARY

Firmware Engineer with 5 years of experience in embedded systems and real-time firmware development. Skilled in STM32, ARM Cortex-M, RTOS (FreeRTOS), communication protocols and hardware interfaces. Proficient in Embedded C/C++ with expertise in sensor interfacing, motor control, and IoT integration. Experienced in Agile development, version control (Git), and modular firmware architecture. Passionate about building scalable, production-ready solutions across biomedical and industrial domains.

WORK EXPERIENCE

Hughes Systique Pvt. Ltd

Bangalore

Senior Engineer

09/2025 - Present

- Worked on the Bare-metal application on the AMD's zynq ultrascale+ MPSoC hardware and the secure boot

Sacra Systems Private Limited

Coimbatore

Firmware Engineer

08/2022 - 07/2025

- Gathered system requirements and designed scalable firmware and system architecture
- Developed and implemented embedded code for complex systems involving real-time constraints
- Collaborated with cross-functional teams to ensure seamless hardware-software integration
- Conducted rigorous testing and debugging to ensure reliability and performance

Associate Firmware Engineer

- Developed embedded firmware for medical devices, BLDC motor controllers, and IoT-based systems
- Resolved critical hardware-to-firmware integration issues, improving system stability and performance
- Collaborated with cross-functional teams to deliver production-ready solutions

Zum Heilen Healthcare Private Limited

Trichur

Hardware Design Engineer

03/2021 - 12/2021

- Contributed to the full lifecycle development of a biomedical product

Aiva Healthcare Technologies Pvt. Ltd

Coimbatore

Hardware Product Developer

04/2020 - 02/2021

- Contributed to the complete product development lifecycle of two innovative devices: an asthma monitoring device and a physiotherapy glove device for stroke patients

PROJECTS

Rotational Disk Angle Measurement Device

Developed a dual-disk angular measurement system to independently track vertical and horizontal rotation using sensor fusion and real-time processing.

- Programmed in **Embedded C** on **STM32**, using **FreeRTOS** to implement an event-driven architecture with dedicated tasks for data acquisition, processing, and communication
- Measured vertical rotation using an IMU sensor interfaced via **SPI**, and horizontal rotation using an optical encoder communicating over **Modbus-RTU**
- Processed sensor data in real time to compute angular displacement for both axes and transmitted results wirelessly via Bluetooth for external visualization
- Followed **Agile development** practices with iterative sprints and task tracking, and Managed codebase using **Git** for version control and team collaboration

IoT Gateway System

- Developed a **C++** application on Ubuntu 24.04 to collect live, event, and health data from IoT nodes via **Modbus-RTU**, process it into structured insights, and transmit it to the cloud using **MQTT**
- Designed a modular, event-driven microservice architecture using OODA principles, and used the **CMake** build system to manage cross-platform builds and modular code organization
- Followed Agile development practices with iterative sprints and task tracking, and used Git for version control and collaborative development

Wearable Asthma Monitoring IoT Device

Developed iHale, a smart wearable device for real-time monitoring of asthma triggers and respiratory distress.

- Programmed in **Bare-metal Embedded C** on **STM32** to collect data from a TI respiratory sensor (via **SPI**), MPU6050 IMU (via **I2C**), and NTC thermistor (via **ADC**)
- Processed multi-sensor data and Transmitted data to the cloud via ESP32 Wi-Fi module for remote access

Smart Glove for Stroke Patients

Designed and developed a wearable glove to assist stroke patients in physiotherapy by tracking precise finger movements.

- Programmed in **Bare-metal Embedded C** on **STM32**, interfacing with 16 MPU6050 IMU sensors via I2C to capture real-time motion data from each finger joint
- Calculated joint angles and transmitted data wirelessly via Bluetooth (HC-05) to a user interface for physiotherapy tracking and visual feedback

Ethernet MQTT Communication for Monitoring Device

Designed and implemented a reliable Ethernet-based communication system for an automotive monitoring device that collects device parameters and transmits them to a remote station for real-time monitoring.

- Integrated **LWIP (Lightweight TCP/IP stack)** on an **STM32** microcontroller running **FreeRTOS** to handle TCP/IP networking efficiently
- Configured and optimized the STM32 to collect sensor data and publish it over **MQTT** running on top of LWIP, ensuring seamless, stable, and reliable data transmission through Ethernet

Motor Controller Development

Developed and optimized a **BLDC motor** control system using **STM32** and STSPIN hardware for precise speed and torque control.

- Implemented **Field-Oriented Control (FOC)** using **STM32 Motor Control SDK (MCSDK)** for smooth and efficient **BLDC motor** operation, enabling closed-loop control for both speed and torque
- Programmed in **Embedded C**, integrating real-time control loops and tuning dynamic parameters to ensure stable performance under varying load conditions